

LONGVILLE, MN, USA

WMO: 727497

Lat: 46.990N

Lon: 94.204W

Elev: 407

StdP: 96.53

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04984

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.1	-27.4	-34.0	0.2	-31.1	-31.2	0.2	-26.4	8.1	-8.7	7.3	-9.3	0.7	300	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	29.0	21.7	27.7	20.7	27.0	20.0	23.6	27.8	22.3	26.2	21.1	24.8	3.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.3	17.9	26.5	21.2	16.6	25.2	19.8	15.2	23.8	72.1	27.8	67.7	25.7	63.2	25.0	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.7	5.6	-34.8	32.1	3.8	1.2	-37.5	33.0	-39.7	33.7	-41.8	34.4	-44.6	35.3		
(5)			-34.9	25.4	3.6	1.2	-37.5	26.2	-39.6	26.9	-41.6	27.6	-44.3	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.5	-12.7	-11.0	-3.3	4.6	11.3	16.6	19.4	17.9	13.9	6.3	-1.3	-9.2
	DBStd	12.54	8.07	7.30	7.44	5.37	4.72	3.48	3.09	3.31	4.63	4.72	5.96	7.11
	HDD10.0	3007	703	589	414	174	39	1	0	1	15	135	339	596
	HDD18.3	5198	961	823	671	413	223	73	24	48	148	372	588	854
	CDD10.0	983	0	0	2	12	81	198	292	246	131	22	1	0
	CDD18.3	133	0	0	0	0	6	20	58	36	13	1	0	0
	CDH23.3	1358	0	0	0	5	104	227	556	334	121	11	0	0
	CDH26.7	274	0	0	0	0	23	39	121	67	22	2	0	0
Wind	WSAvg	2.2	2.2	2.2	2.3	2.4	2.3	2.0	1.8	1.7	2.2	2.5	2.5	2.1
Precipitation	PrecAvg	710	18	16	34	54	81	109	104	79	80	79	33	22
	PrecMax	855	49	49	89	131	192	190	193	160	180	152	104	54
	PrecMin	529	2	1	9	21	31	37	48	22	19	10	4	7
	PrecStd	85	13	11	22	25	37	40	40	39	34	40	24	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	7.9	17.8	23.0	29.0	29.8	32.2	31.2	29.1	24.2	17.3	6.3
		MCWB	3.5	3.9	11.7	13.7	18.3	22.4	24.5	23.0	20.6	15.6	11.1	3.2
	2%	DB	2.5	4.1	12.9	19.8	26.3	27.6	29.1	28.0	27.0	20.9	12.9	3.7
		MCWB	0.7	1.6	8.2	11.5	16.0	20.3	22.3	21.5	20.3	14.0	9.0	2.0
	5%	DB	0.3	2.3	9.1	17.4	23.0	26.2	27.8	27.2	24.0	17.4	10.0	2.0
		MCWB	-0.6	0.5	5.5	9.1	14.2	18.6	21.2	20.5	18.6	12.2	7.0	0.8
	10%	DB	-2.0	0.0	7.0	14.2	21.1	24.0	27.0	25.1	22.4	13.9	7.4	0.1
		MCWB	-2.7	-1.1	3.8	7.6	13.6	17.5	20.6	19.2	17.8	9.7	4.8	-0.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	4.7	14.1	14.8	20.7	24.1	25.4	24.2	23.1	17.2	12.7	4.1
		MCDB	6.6	8.2	16.5	22.0	26.7	28.3	29.6	28.7	26.7	20.7	14.9	5.0
	2%	WB	1.1	2.1	8.8	12.0	18.1	21.9	23.9	22.9	21.3	15.0	9.6	2.1
		MCDB	2.5	4.0	12.1	17.5	22.5	25.9	28.3	26.8	24.9	19.7	13.2	3.6
	5%	WB	-0.7	0.4	6.1	10.1	16.4	20.3	22.6	21.6	19.7	12.7	7.1	0.7
		MCDB	0.2	2.1	9.3	16.1	20.7	24.2	26.7	25.5	23.5	15.9	9.4	1.4
	10%	WB	-2.4	-1.2	3.8	8.3	14.7	18.9	21.3	20.4	18.4	10.8	4.8	-0.6
		MCDB	-2.0	-0.1	6.3	13.7	19.5	22.5	24.9	24.0	21.6	13.6	7.4	0.0
Mean Daily Temperature Range	5% DB	MDBR	10.8	12.0	11.9	12.3	12.9	11.9	12.0	12.1	12.0	9.8	8.8	8.8
		MCDBR	10.8	12.3	14.5	17.6	17.8	14.2	13.4	13.7	14.2	15.3	13.5	9.7
		MCWBR	8.9	9.6	9.5	10.0	8.8	7.6	7.4	7.5	8.4	8.9	9.0	7.9
	5% WB	MCDBR	10.2	11.2	13.4	15.8	14.1	12.0	11.9	11.9	12.2	13.5	12.5	8.7
		MCWBR	8.6	8.7	9.2	9.6	8.1	7.4	7.4	7.0	7.7	8.4	9.1	7.5
Clear-Sky Solar Irradiance	taub	0.255	0.280	0.307	0.352	0.370	0.372	0.379	0.381	0.352	0.309	0.282	0.263	
	taud	2.369	2.302	2.388	2.360	2.367	2.389	2.387	2.399	2.459	2.572	2.518	2.394	
	Ebn at Noon	850	892	921	902	894	892	880	863	860	859	817	795	
	Edh at Noon	69	93	100	114	118	116	115	109	94	72	61	61	
All-Sky Solar Radiation	RadAvg	1.56	2.69	3.81	4.78	5.34	5.80	6.19	5.21	3.87	2.37	1.43	1.09	
	RadStd	0.17	0.26	0.38	0.43	0.37	0.40	0.27	0.42	0.29	0.35	0.15	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (45 neighbors)	N/A	-2.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page